Project: DeskBot

A small desktop companion robot with:

- 🤖 Animated face (eyes, expressions)
- 🎤 Voice commands
- 📱 Mobile app controls
- 🌤️ Weather/time display
- 📚 Study timer (Pomodoro)
- 🔔 Salah reminders
- 🎵 Music controls
- 💬 AI chat mode
- 🦾 Head movement and arm movement

Version 1 (Build this first)

Hardware:

- ESP32
- OLED screen
- 2 servo motors
- Small speaker
- Battery pack

Software:

- React Native app
- Bluetooth connection
- Face animations
- Basic controls

Features:

- Move head left/right
- Show emotions
- Display weather
- Study timer

This is realistic to build in a few weeks.

Version 2

Add:

- Voice recognition
- Wake word
- ChatGPT API
- Camera
- Face tracking

Now it starts looking like a real AI desk companion.

Tech Stack

For you specifically:

Frontend App:

- React Native

Backend:

- Node.js

Robot Controller:

- ESP32 by Espressif

Design:

- Affinity Designer

CAD:

- Onshape

Week 1:

- CAD design
- Robot body
- 3D printing

Week 2:

- ESP32
- Servos
- OLED face

Week 3:

- Bluetooth controls

Week 4:

- Mobile app

Week 5:

- AI features

Week 6:

- Polish and publish

Publish:

- GitHub repo
- Build log
- Demo video
- Documentation

Start with:

1. Design the robot in CAD.
2. Build the mobile app interface.
3. Create the animated face.
4. Log everything through Hack Club/Hackatime.
5. Add hardware later when you get parts.